Symplectic Euler Integration

From Hamiltonian Mechanics to Real-Time Physics Simulation

Ernest Yeung

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1 Introduction

Physics engines such as NVIDIA PhysX (used inside Isaac Sim) must integrate Newton's equations of motion millions of times per second at a fixed timestep h. Naive application of the simplest numerical integrator—explicit (forward) Euler—causes energy to grow without bound, producing bodies that accelerate spontaneously. The standard remedy is the *symplectic Euler* (also called *semi-implicit Euler*) method, which is the unique first-order scheme that exactly preserves the geometric structure underlying Hamiltonian mechanics.

This note derives that structure from first principles, proves that explicit Euler breaks it while symplectic Euler preserves it, connects the method to the classical generating-function theory of canonical transformations, and works out the exact discrete free-fall formula that emerges—the formula used to validate the PhysX integrator in the Starship IsaacSim test suite (see `tests/test_t1_free_fall.py`).

2 Hamiltonian Mechanics and the Symplectic Form

2.1 Phase Space and Hamilton's Equations

Let $q \in \mathbb{R}^n$ denote generalised coordinates and $p \in \mathbb{R}^n$ the conjugate momenta. The pair (q, p) lives in the *phase space* $T^*Q \cong \mathbb{R}^{2n}$. Given a Hamiltonian $H : T^*Q \to \mathbb{R}$, the equations of motion are

$$\dot{q} = \frac{\partial H}{\partial p}, \qquad \dot{p} = -\frac{\partial H}{\partial q}. \tag{1}$$

These can be written compactly as $\dot{z} = \mathbf{\Omega}^{-1}\nabla H(z)$, where $z = (q, p)^\top$ and

$$\mathbf{\Omega} = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix} \in \mathbb{R}^{2n\times 2n} \tag{2}$$

is the *standard symplectic matrix* (satisfying $\mathbf{\Omega}^\top = -\mathbf{\Omega}$, $\mathbf{\Omega}^2 = -I$).

2.2 The Symplectic 2-Form

On T^*Q there is a canonical closed, non-degenerate 2-form

$$\omega = \sum_{i=1}^{n} \mathrm{d}q_i \wedge \mathrm{d}p_i. \tag{3}$$

In coordinates, a linear map $\Phi : \mathbb{R}^{2n} \to \mathbb{R}^{2n}$ with matrix M is called *symplectic* if it preserves ω:

$$M^\top \mathbf{\Omega}\, M = \mathbf{\Omega}. \tag{4}$$

For $n = 1$ this reduces to $\det(M) = 1$, i.e., the map preserves oriented area in the (q, p)-plane.

2.3 Liouville's Theorem

Theorem 2.1 (Liouville)**.** *The flow φ_t of a Hamiltonian system (1) is symplectic for all t: $(\varphi_t)^*\omega = \omega$. In particular it preserves the $2n$-dimensional phase-space volume $\mathrm{d}q\,\mathrm{d}p$.*

Sketch. Differentiating $M(t) = D\varphi_t(z_0)$ along the flow and using $\dot{M} = JM$ where $J = \mathbf{\Omega}^{-1}D^2H$ is the linearised vector field gives $\frac{\mathrm{d}}{\mathrm{d}t}(M^\top\mathbf{\Omega}M) = 0$, so $M^\top\mathbf{\Omega}M$ is constant. At $t = 0$, $M(0) = I$ and $I^\top\mathbf{\Omega}I = \mathbf{\Omega}$, proving the claim for all t. □

3 Explicit Euler: Not Symplectic

3.1 The Algorithm

For a particle of mass m with Hamiltonian $H = \frac{p^2}{2m} + V(q)$ (one degree of freedom, $n = 1$), Hamilton's equations are

$$\dot{q} = \frac{p}{m}, \qquad \dot{p} = F(q) := -V'(q). \tag{5}$$

The **explicit (forward) Euler** discretisation uses state values from step n to advance to step $n + 1$:

$$p_{n+1} = p_n + h\,F(q_n), \tag{6}$$

$$q_{n+1} = q_n + \frac{h}{m}\,p_n. \tag{7}$$

Both updates use the *old* values (q_n, p_n).

3.2 Jacobian and Failure of Symplecticity

The Jacobian of the map $(q_n, p_n) \mapsto (q_{n+1}, p_{n+1})$ is

$$\mathbf{J}_{\text{expl}} = \begin{pmatrix} \partial q_{n+1}/\partial q_n & \partial q_{n+1}/\partial p_n \\ \partial p_{n+1}/\partial q_n & \partial p_{n+1}/\partial p_n \end{pmatrix} = \begin{pmatrix} 1 & h/m \\ hF'(q_n) & 1 \end{pmatrix}. \tag{8}$$

Its determinant is

$$\det(\mathbf{J}_{\text{expl}}) = 1 - \frac{h^2}{m} F'(q_n). \tag{9}$$

For any non-trivial potential ($F' \neq 0$) this differs from 1, so explicit Euler *does not* satisfy the symplectic condition (4).

Consequence. For a harmonic oscillator with $F(q) = -kq$ (spring constant $k > 0$), $F'(q) = -k$, giving

$$\det(\mathbf{J}_{\text{expl}}) = 1 + \frac{h^2 k}{m} > 1.$$

Each step *expands* phase-space area by the factor $1 + h^2k/m > 1$, which compounds over N steps to $(1 + h^2k/m)^N \to \infty$. This corresponds to spurious energy growth that makes explicit Euler numerically unstable for oscillatory systems at any fixed $h > 0$.

4 Symplectic Euler: Preserving the Structure

4.1 The Algorithm

The **symplectic (semi-implicit) Euler** method updates momentum first using the old position, then updates position using the *new* momentum:

$$p_{n+1} = p_n + h\,F(q_n), \tag{10}$$

$$q_{n+1} = q_n + \frac{h}{m}\,p_{n+1}. \tag{11}$$

The only difference from (6)–(7) is that (11) uses p_{n+1} rather than p_n.

4.2 Jacobian and Proof of Symplecticity

Substituting (10) into (11):

$$q_{n+1} = q_n + \frac{h}{m}(p_n + hF(q_n)) = q_n + \frac{h}{m}p_n + \frac{h^2}{m}F(q_n).$$

The Jacobian is therefore

$$\mathbf{J}_{\text{symp}} = \begin{pmatrix} 1 + \frac{h^2}{m}F'(q_n) & \frac{h}{m} \\ hF'(q_n) & 1 \end{pmatrix}. \tag{12}$$

Proposition 4.1. *The map* (10)–(11) *is symplectic, i.e.,* $\mathbf{J}_{\text{symp}}^\top \boldsymbol{\Omega}\, \mathbf{J}_{\text{symp}} = \boldsymbol{\Omega}$ *(equivalently,* $\det(\mathbf{J}_{\text{symp}}) = 1$*).*

Proof. Let $\alpha = h^2F'(q_n)/m$, $\beta = h/m$, $\gamma = hF'(q_n)$, so $\alpha = \beta\gamma$. Then

$$\mathbf{J}_{\text{symp}} = \begin{pmatrix} 1+\alpha & \beta \\ \gamma & 1 \end{pmatrix}.$$

We verify the symplectic identity with $\boldsymbol{\Omega} = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$:

$$\mathbf{J}^\top \boldsymbol{\Omega} \mathbf{J} = \begin{pmatrix} 1+\alpha & \gamma \\ \beta & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 1+\alpha & \beta \\ \gamma & 1 \end{pmatrix}.$$

First factor on the right:

$$\begin{pmatrix} 1+\alpha & \gamma \\ \beta & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = \begin{pmatrix} -\gamma & 1+\alpha \\ -1 & \beta \end{pmatrix}.$$

Then:

$$\begin{aligned} \mathbf{J}^\top \boldsymbol{\Omega} \mathbf{J} &= \begin{pmatrix} -\gamma & 1+\alpha \\ -1 & \beta \end{pmatrix} \begin{pmatrix} 1+\alpha & \beta \\ \gamma & 1 \end{pmatrix} \\ &= \begin{pmatrix} -\gamma(1+\alpha) + (1+\alpha)\gamma & -\gamma\beta + (1+\alpha) \\ -(1+\alpha) + \beta\gamma & -\beta + \beta \end{pmatrix} \\ &= \begin{pmatrix} 0 & 1+\alpha-\gamma\beta \\ \beta\gamma - 1 - \alpha & 0 \end{pmatrix}. \end{aligned}$$

Since $\alpha = \beta\gamma$:

$$1 + \alpha - \gamma\beta = 1 + \beta\gamma - \gamma\beta = 1, \quad \beta\gamma - 1 - \alpha = \beta\gamma - 1 - \beta\gamma = -1.$$

Therefore $\mathbf{J}^\top \boldsymbol{\Omega} \mathbf{J} = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right) = \boldsymbol{\Omega}$. □ □

The key cancellation $\alpha = \beta\gamma$ is the algebraic fingerprint of using p_{n+1} in the position update: it makes the off-diagonal contributions to det cancel exactly.

5 Derivation from the Generating-Function Theory

5.1 Type-2 Generating Functions

Canonical transformation theory [3] classifies symplectomorphisms by the type of mixed-variable generating function S_k used to define them. A *Type-2 generating function* $S_2(q, P)$ of old position q and new momentum P generates the transformation via

$$p = \frac{\partial S_2}{\partial q}, \qquad Q = \frac{\partial S_2}{\partial P}, \tag{13}$$

and any S_2 satisfying these relations produces a canonical (symplectic) map $(q, p) \mapsto (Q, P)$ by construction [1].

5.2 Symplectic Euler from a Type-2 Function

Choose the generating function

$$S_2(q_n, p_{n+1}) = q_n \cdot p_{n+1} + h\, H(q_n, p_{n+1}). \tag{14}$$

For $H = p^2/(2m) + V(q)$ this becomes

$$S_2(q_n, p_{n+1}) = q_n\, p_{n+1} + h\left[\frac{p_{n+1}^2}{2m} + V(q_n)\right].$$

Applying (13) with $q \leftarrow q_n$, $P \leftarrow p_{n+1}$, $p \leftarrow p_n$, $Q \leftarrow q_{n+1}$:

$$p_n = \frac{\partial S_2}{\partial q_n} = p_{n+1} + h\, V'(q_n) = p_{n+1} - h\, F(q_n), \tag{15}$$

$$q_{n+1} = \frac{\partial S_2}{\partial p_{n+1}} = q_n + \frac{h}{m}\, p_{n+1}. \tag{16}$$

Solving (15) for p_{n+1}:

$$p_{n+1} = p_n + h\, F(q_n),$$

which is exactly (10). Substituting into (16) recovers (11). Since every transformation generated by a Type-2 function is canonical (it satisfies (4) by construction of the generating-function formalism), this provides an independent proof of proposition 4.1 and reveals the *origin* of the method: symplectic Euler is the degree-1 truncation in h of the exact flow map, approximated via S_2.

Remark 5.1. *The identity $S_2(q, P) = qP$ generates the identity transformation. Adding $hH(q, P)$ as a correction gives a first-order approximation to the exact flow, but one that* exactly *preserves the symplectic structure at every $h > 0$.*

6 Modified Hamiltonian and Bounded Energy Error

A symplectic integrator does not conserve H exactly; instead it exactly conserves a *modified Hamiltonian* $\tilde{H}$ close to H [1, 2]:

$$\tilde{H}(q, p) = H(q, p) + \frac{h}{2m}\, p\, V'(q) + \mathcal{O}(h^2). \tag{17}$$

This is the key distinction from non-symplectic methods:

- **Explicit Euler**: energy error grows *monotonically* as $(1 + h^2 k/m)^N$ — unbounded.
- **Symplectic Euler**: energy error oscillates around a constant of order $\mathcal{O}(h)$, bounded for all time.

The $\mathcal{O}(h)$ offset between H and $\tilde{H}$ is why T1 sees an energy drift of 6×10^{-4} (bounded, not growing).

7 Exact Discrete Solution: Free Fall

7.1 Derivation

For vertical free fall $H = p^2/(2m) + mgz$, $F = -mg$, the symplectic Euler update becomes

$$
\begin{aligned}
v_{n+1} &= v_n - gh, && (18)\\
z_{n+1} &= z_n + h\, v_{n+1} = z_n + h(v_n - gh). && (19)
\end{aligned}
$$

where $v_n = p_n/m$. Telescoping (18):

$$v_n = v_0 - ngh. \tag{20}$$

Telescoping (19):

$$
\begin{aligned}
z_n &= z_0 + h\sum_{k=1}^{n} v_k = z_0 + h\sum_{k=1}^{n}(v_0 - kgh)\\
&= z_0 + nv_0 h - gh^2 \sum_{k=1}^{n} k = z_0 + nv_0 h - gh^2 \cdot \frac{n(n+1)}{2}. && (21)
\end{aligned}
$$

Setting $t = nh$ so that $n = t/h$:

$$\boxed{z(t) = z_0 + v_0 t - \tfrac{1}{2} g\, t\,(t+h).} \tag{22}$$

7.2 Comparison with Continuous Theory

The continuous solution is $z_{\text{cont}}(t) = z_0 + v_0 t - \frac{1}{2}gt^2$. Expanding (22):

$$z(t) = z_{\text{cont}}(t) - \frac{gh}{2}\, t.$$

The discrete solution lags the continuous one by $\frac{1}{2}ght$, which is $\mathcal{O}(h)$ at each instant—confirming first-order accuracy. At $t = 3\,\text{s}$ and $h = 1/240\,\text{s}$ (PhysX default), the lag is $\frac{1}{2} \times 9.81 \times \frac{1}{240} \times 3 \approx 0.061\,\text{m}$, which exceeds the test tolerance of $0.01\,\text{m}$ if the wrong (continuous) formula is used for comparison. Using (22) reduces the residual to $< 10^{-4}\,\text{m}$ (the test observes $0.0001\,\text{m}$).

	Position formula	Area-preserving?
Explicit Euler	$z_0 + v_0 t - \frac{1}{2}gt^2$	No
Symplectic Euler	$z_0 + v_0 t - \frac{1}{2}g\,t(t+h)$	Yes
Continuous	$z_0 + v_0 t - \frac{1}{2}gt^2$	(exact)

Table 1: Free-fall position formulas for $n = t/h$ steps.

8 Summary

The chain of equivalences is:

$$\underbrace{\text{Type-2 generating function}}_{\text{canonical transformation}} \Rightarrow \underbrace{\text{symplectic map}}_{\mathbf{J}^\top\mathbf{\Omega}\mathbf{J}=\mathbf{\Omega}} \Rightarrow \underbrace{\text{area-preserving}}_{\det\mathbf{J}=1} \Rightarrow \underbrace{\text{bounded energy error}}_{\tilde{H}=\text{const}}.$$

The one-line algorithmic change—replace p_n with p_{n+1} in the position update—makes $\alpha = \beta\gamma$ cancel, flips an expanding map to an area-preserving one, and gives PhysX the energy stability it needs for real-time simulation.

References

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